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Age-Specific Cerebral Vessel Templates Across the Lifespan of Healthy Adults

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Chang Xu, Ziyu Meng, Huixiang Zhuang, Ruihao Liu, Li Cao, Zhi-Pei Liang, Yaou Liu & Yao Li

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Title

Age-Specific Cerebral Vessel Templates Across the Lifespan of Healthy Adults

Authors

Chang Xu[1], Ziyu Meng[1,2], Huixiang Zhuang[1], Ruihao Liu[3], Li Cao[4], Zhi-Pei Liang[3,5], Yaou Liu[6,7], and Yao Li[1,2,8]

Affiliations

1. National Engineering Research Center of Advanced Magnetic Resonance Technologies for Diagnosis and Therapy (NERC-AMRT), School of Biomedical Engineering, Shanghai Jiao Tong University, Shanghai, China.
2. Institute of Medical Robotics, Shanghai Jiao Tong University, Shanghai, China.
3. Beckman Institute for Advanced Science and Technology, University of Illinois at Urbana-Champaign, Urbana, IL, United States.
4. Department of Neurology, and Shanghai Neurological Rare Disease Biobank and Precision Diagnostic Technical Service Platform, Shanghai Sixth People's Hospital Affiliated to Shanghai Jiao Tong University School of Medicine, Shanghai, China.
5. Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign, Urbana, IL, United States.
6. Department of Radiology, Beijing Tiantan Hospital, Capital Medical University, Beijing, China.
7. Tiantan Image Research Center, China National Clinical Research Center for Neurological Diseases, Beijing, China
8. Shanghai Center for Brain Science and Brain-Inspired Technology, Shanghai, China.

Corresponding author: Yao Li (yaoli@sjtu.edu.cn) and Ziyu Meng (mengzy0306@sjtu.edu.cn)

Abstract

Cerebral vasculature plays a critical role in brain function. Accurate characterization of its normative organization and distribution is essential for identifying vascular abnormalities. However, existing cerebrovascular templates do not adequately account for age-related anatomical variability and typically rely on structural-image-driven registration without explicitly leveraging vascular information for alignment. To address these limitations, we present a set of open, age-specific cerebrovascular templates constructed from time-of-flight magnetic resonance angiography (TOF-MRA) data of 1,288 healthy adults spanning the adult lifespan (19-92 years). Fine-scale vascular structures were automatically segmented using CereVessPro and registered through a novel two-stage framework that integrates age-specific structural alignment with vessel-guided refinement. The resulting resources comprise a series of openly accessible maps, including age-specific MRA templates, vessel probability maps, vessel density maps, and vessel radius maps from young to older adulthood. This framework is designed to improve vascular correspondence across subjects by explicitly addressing age-dependent anatomical differences and incorporating vascular features during alignment. Validation analyses revealed biologically plausible age-related changes in vascular morphology and vessel calibre, particularly in major cerebral arteries. Together, these data provide an age-resolved reference for cerebrovascular mapping, improving vascular alignment and supporting more reliable group-level analyses. This resource may facilitate more sensitive detection of vascular abnormalities in both research and clinical contexts.

Background & Summary

The cerebral vasculature is essential for sustaining brain function by delivering oxygen and nutrients to support neuronal activity and maintain metabolic homeostasis. Structural or functional abnormalities in cerebral vessels contribute to a range of cerebrovascular diseases, including aneurysms[1], stenosis[2], small vessel disease (SVD)[3] and stroke[4]. Vascular compromise also plays a significant role in neurodegenerative

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diseases[5]; cerebral hypoperfusion and microvascular degeneration are increasingly implicated in dementias pathogenesis[6-8]. Therefore, early detection of such abnormalities is critical, as many lead to irreversible tissue damage and long-term neurological deficits. Establishing normative references of vascular structure across healthy populations provides an essential foundation for identifying subtle vascular alterations and enabling early detection and diagnosis of cerebrovascular and neurodegenerative disorders.

Cerebral vascular architecture is not static but undergoes continuous remodeling across the human lifespan. Aging is accompanied by progressive vascular degeneration, reflected in endothelial dysfunction, arterial stiffening, inflammation, reduced cerebrovascular reactivity, brain-blood barrier (BBB) leakage, and microvasculature rarefaction[9-12]. These physiological changes manifest in imaging features such as reduced vascular volume[13], altered vessel diameter[14], and increased tortuosity[15]. Age-related decline in collateral circulation, particularly within the circle of Willis, have also been reported, with posterior communicating arteries becoming less prominent or absent in older adults[14,16,17]. Therefore, a single age-agnostic vascular template cannot fully capture the normative diversity of cerebrovascular anatomy, highlighting the need for age-specific references to support sensitive detection of vascular abnormalities in clinical and research applications.

Several cerebral vascular templates have been developed over the past decade. Early efforts were based on small sample sizes in narrow age ranges[18,19], limiting their generalizability across the lifespan. More recent studies expanded cohort sizes and age coverage[20,21]. For instance, Mouches et al.[20] released a public arterial atlas derived from magnetic resonance angiography (MRA) data of 544 subjects aged 19-86 years using nonlinear registration. Wang et al.[21] constructed a cerebrovascular atlas from 1336 adults aged 19-92 years. Despite these advances, most existing approaches rely on a single pooled template and perform registration primarily using T1-weighted (T1w) images, without explicitly incorporating vascular features as alignment constraints. This may limit the accuracy of vessel correspondence, particularly for small or tortuous vessels. Prior work has shown that incorporating vascular information can improve registration accuracy[22,23]. In addition, constructing a single atlas across a wide age range may obscure systematic age-related anatomical differences, potentially reducing alignment fidelity across heterogeneous populations.

In this study, we constructed a set of open, age-specific cerebrovascular templates based on TOF-MRA data of 1,288 healthy adults spanning the adult lifespan (19-92 years). To capture fine-scale vascular morphology, we applied CereVessPro, a state-of-the-art vessel segmentation methodology trained on large-scale, high-quality cerebrovascular annotations[13]. We used a two-stage registration method to integrate global structural alignment with vessel-level refinement. By first aligning individual brains to age-specific structural templates, this registration method effectively accounted for age-dependent anatomical variations; subsequent incorporation of vascular features enabled accurate matching of fine-scale vascular structures. The resulting age-specific maps may serve as age-resolved references for characterizing vascular variation across the lifespan, enabling detailed analysis of cerebrovascular organization and supporting more reliable group-level comparisons.

Methods

Datasets

A total of 1337 healthy subjects with paired T1w MRI and TOF-MRA data were collected from five public databases. For individuals with multiple imaging sessions, only the baseline scan was included. Together, these datasets provided broad lifespan coverage, multi-site variability, and diverse acquisition protocols.

To capture vascular and structural characteristics across early to late adulthood, we combined datasets with extensive age ranges and whole brain coverage. The MIDAS database[22], released by CASILab at the University of North Carolina at Chapel Hill (UNC) (https://data.kitware.com/#collection/591086ee8d777f16d01e0724), is publicly accessible via the Kitware Data Server, and provided paired T1w-MRA scans from 109 healthy volunteers aged 19-72 years, excluding diabetes, hypertension, head trauma, psychiatric disease, or other conditions may influence brain structure. Similarly, the International Consortium for Brain Mapping (ICBM) database[23] (https://ida.loni.usc.edu/), accessed via the LONI Image Data Archive with registration and approval under a Data Use Agreement, provided 189 T1w-MRA pairs from individuals aged 19-80 years collected at three institutions, Montreal Neurologic Institute (ICBM-MNI), the University of California, Los Angeles (UCLA), and the University of Texas Health Science Center at San Antonio (UTHC), representing a racially and ethnically diverse healthy population with no history of stroke.

To enhance representation in young adults and incorporate data with repeated high-quality imaging sessions, we included the Midnight Scan Club (MSC) database[24] (https://openneuro.org/datasets/ds000224/versions/1.0.3), a publicly available database on OpenNeuro. One baseline scan from each of ten participants aged 24-34 years was selected. To further expand the overall sample size and age diversity, two large-scale repositories were incorporated. The Information eXtraction from Images (IXI) database[25] (https://brain-development.org/ixi-dataset/) is publicly accessible via the official repository and contributed 569 datasets from participants aged 20-86 years, acquired at three hospitals in London (Hammersmith Hospital, Institute of Psychiatry, and Guy's Hospital). The Open Access Series of Imaging Studies (OASIS)-3 database[26] (http://www.oasis-brains.org), accessed via the official website and requires a standard Data Use Agreement, provided 460 baseline scans from cognitively normal older adults aged 43-92 years, where cognitive normality was defined by a Clinical Dementia Rating (CDR) of zero. Across datasets, "healthy" was operationally defined based on the inclusion and exclusion criteria provided by each source, which primarily focus on the absence of major neurological or cognitive disorders. Detailed information on cardiovascular risk factors (e.g., hypertension, diabetes, or smoking status) is not consistently available across all datasets. As a result, individuals with unreported or mild cardiovascular risk factors may be included. While this heterogeneity may introduce additional variability in the derived measures, it may also better reflect real-world population diversity across the adult lifespan. Across datasets, "healthy" was operationally defined based on the inclusion and exclusion criteria provided by each source, which primarily focus on the absence of major neurological or cognitive disorders. Detailed information on cardiovascular risk factors (e.g., hypertension, diabetes, or smoking status) is not consistently available across all datasets. As a result, individuals with unreported or mild cardiovascular risk factors may be included. While this heterogeneity may introduce additional variability in the derived measures, it may also better reflect real-world population diversity across the adult lifespan.

After standardized preprocessing and rigorous quality control (details below), 1288 datasets were included for analysis. Demographic information and acquisition parameters are summarized in Table 1 and Table 2, respectively. Subjects were categorized into four age groups following the United Nations' age group definitions[27-29]: (1) youth (YO, 19-24 years old, N=74); (2) young adults (YA, 25-44 years old, N=332); (3) middle-aged adults (MA, 45-64 years old, N=415); (4) older adults (OA, 65+years old, N=467). Subjects aged below 19 years were not included due to the lack of publicly available datasets. Detailed demographic characteristics and sample distributions for each group are provided in Table 3 and Table 4, respectively.

Table 1. Demographic information of all included subjects.

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Dataset	Number of Subjects	Range of Age (years)	Sex		Mean Age (years)		
			Male	Female	Male	Female	Total
MIDAS	95	19-72	43	52	40.98 ± 13.71	44.21 ± 14.67	42.75 ± 14.26
ICBM	184	19-80	83	101	39.77 ± 13.62	43.79 ± 16.64	41.98 ± 15.44
MSC	10	24-34	5	5	30.40 ± 3.78	27.80 ± 2.59	29.10 ± 3.35
IXI	544	20-86	241	303	46.58 ± 16.47	50.20 ± 16.50	48.60 ± 16.57
OASIS-3	455	43-92	198	257	70.47 ± 8.28	69.08 ± 8.15	69.68 ± 8.23
Total	1288	19-92	570	718	53.32 ± 18.53	55.47 ± 17.44	54.52 ± 17.96

Table 2. TOF-MRA acquisition parameters. TR: repetition time; TE: echo time; FA: flip angle; FOV: field of view.

Dataset	Manufacturer	Model	Field Strength (T)	Resolution (mm^3)	TR/TE (ms)	FA (°)	Foot-Head FOV (mm)
MIDAS	Siemens	Allegra	3	0.51×0.51×0.8	35/3.56	22	102.4-140.8
ICBM-MNI	Siemens	Sonata Vision	1.5	0.62×0.62×0.6	33/6.9	25	104.4-148.2
ICBM-UCLA	Siemens	Allegra	3	0.62×0.62×0.62	24/4.3	18	115.94
ICBM-UTHC	Siemens	TIM Trio	3	0.62×0.62×0.62	24/4.3	18	123.38
MSC	Siemens	TIM Trio	3	0.625×0.625×1	25/3.34	20	268
IXI-HH	Philips	Intera	3	0.47×0.47×0.8	16.7/5.75	16	80
IXI-IOP	GE	/	1.5	0.2637×0.2637×0.8	/	/	73.6
IXI-Guys	Philips	Gyroscan Intera	1.5	0.47×0.47×0.8	20/6.9	25	80
OASIS-3	Siemens	TIM Trio	3	0.3×0.3×0.6	23/3.6	18	63.7-152.4

Table 3. Demographic information of each age group. YO: youth; YA: young adults; MA: middle-aged adults; OA: older adults.

Age Group	Number of Subjects	Range of Age (years)	Sex		Mean Age (years)		
			Male	Female	Male	Female	Total
YO	74	19-24	32	42	21.94 ± 1.65	22.15 ± 1.69	22.06 ± 1.67
YA	332	25-44	172	160	33.72 ± 5.54	33.91 ± 6.09	33.81 ± 5.80
MA	415	45-64	165	250	55.61 ± 5.77	56.93 ± 5.40	56.40 ± 5.58
OA	467	>65	201	266	73.21 ± 5.74	72.32 ± 5.45	72.70 ± 5.59
Total	1288	19-92	570	718	53.32 ± 18.53	55.47 ± 17.44	54.52 ± 17.96

Table 4. Distributions of the datasets over the different age groups. YO: youth; YA: young adults; MA: middle-aged adults; OA: older adults.

Dataset	Number of Subjects	YO (19-24 years)	YA (25-44 years)	MA (45-64 years)	OA (>65 years)
MIDAS	95	9	45	33	8
ICBM	184	31	72	70	11
MSC	10	1	9	0	0
IXI	544	33	205	196	110
OASIS-3	455	0	1	116	338

Total	1288	74	332	415	467

Figure 1 shows the cerebrovascular template construction pipeline in this study, including four main steps: (1) data preprocessing and quality control; (2) vessel feature extraction and quality control; (3) two-stage structural and vascular alignment; and (4) age-specific cerebrovascular templates generation.

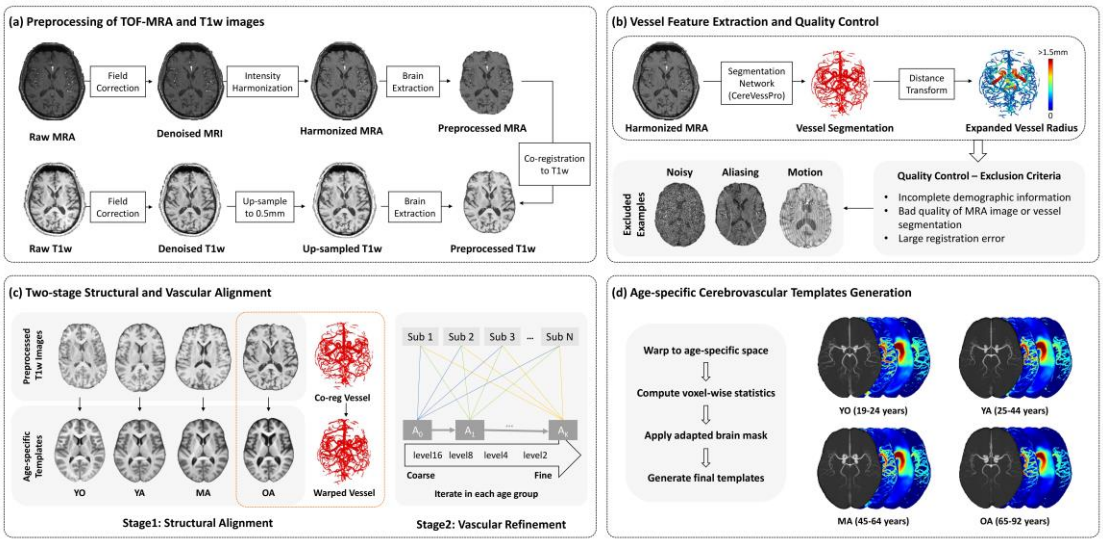


Figure 1. Overall framework of cerebrovascular templates construction. (a) Preprocessing of TOF-MRA and T1w images including intensity harmonization and co-registration. **(b)** Vessel feature extraction using CereVessPro segmentation and radius estimation with quality control exclusions. **(c)** Two-stage registration framework combining structural and vascular alignment. **(d)** Age-specific cerebrovascular templates generation. Abbreviations: YO: youth; YA: young adults; MA: middle-aged adults; OA: older adults.

Preprocessing of TOF-MRA and T1w images

The overall preprocessing pipeline is summarized in Fig. 1a. For TOF-MRA images, N4 bias field correction[30] was first applied using the N4ITK implementation in the Insight Toolkit[31] (ITK, https://itk.org/). Brain extraction was performed with SynthStrip[32] embedded in Freesurfer (https://surfer.nmr.mgh.harvard.edu/). To reduce intensity variations arising from differing acquisition parameters across sites, intensity harmonization was performed. Specifically, one high-quality MRA image was manually selected from each age group as a reference; all reference images were chosen from the same site and acquired using an identical scanning protocol. Remaining MRA images were harmonized to their respective reference following the method described by Mouches et al.[20]. T1w images underwent the same N4 bias field correction[30]. To preserve the native MRA spatial resolution and harmonize the resolutions of multi-center MRA images, an isotropic resolution of 0.5 mm was used for the subsequent analyses, and T1w images were up-sampled to match this resolution prior to registration. Brain extraction for T1w was subsequently performed using SynthStrip[32]. Preprocessed MRA images were then registered to their corresponding up-sampled T1w using an affine transformation.

Vessel feature extraction and quality control

Vessel segmentation was performed using CereVessPro[13], a publicly available state-of-the-art nnUNet-based model for TOF-MRA (https://github.com/YingChen7/HQA_Cerebrovasculature/). The model

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integrates an attention module and a hard-example-mining loss, and was pretrained on large-scale, annotated multi-center datasets using a contrastive learning strategy to improve generalizability. Following segmentation, a vessel density map was generated by propagating the segmented vessels into their surrounding neighbourhood. Specifically, the vessel density map was obtained by convolving the vessel segmentation with a 3D spherical kernel of size [30 30 30] voxels for all subjects follows prior work by Sarabi et al.[33], resulting in a smoothed representation of vascular distribution. Prior to convolution, vessel segmentations were resampled to an isotropic resolution of 0.5 mm. In addition, dural sinuses were excluded using a manually defined population mask warped from standard space to minimize the influence of large venous structures. The resulting vessel density maps enable the comparison of vascular density across population without forcing the precise alignment of fine vascular structures, which often exhibit substantial inter-subject variability in topology and branching patterns[34,35].

Voxel-wise vessel radius estimation was conducted in two stages. First, 3D vessel centerlines were extracted using an established method[36,37] implemented in scikit-image[38]. For each centerline voxel, the vessel radius was defined as its Euclidean distance to the nearest non-vessel voxel computed using a distance transform algorithm[39] implemented in SciPy[40]. A second distance transform operation then propagated these centerline-derived radius values to all voxels within the vessel mask by assigning each non-centerline voxel of its nearest centerline voxel. This process effectively expands the vessel boundaries while preserving centerline-derived radii, improving spatial sampling and reducing sensitivity to small registration errors compared with strategies that rely solely on centerline voxels or simple boundary dilation. In addition, the dural sinuses were also excluded due to their inconsistency across datasets.

A rigorous quality control pipeline was implemented throughout the processing stages. First, datasets missing demographic information or officially reporting adverse health events were removed. Then, all retained datasets underwent systematic visual inspection and were excluded if they exhibited excessive noise, aliasing, motion artifacts, and other significant distortions that could compromise segmentation or registration. Segmentation outputs were visually inspected by overlaying vessel masks on MRA images. Cases with vessel volume or vessel radius exceeding 3 standard deviations from the mean were visually double-checked. Finally, after each subsequent registration step (described below), an additional visual quality control confirmed alignment reliability.

Two-stage alignment

Structural alignment

We first aligned all datasets to age-specific structural templates from our previous work[29] (https://doi.org/10.6084/m9.figshare.28093595.v1), which account for age-related atrophy. Each subject's MRA image and vessel segmentation were registered to the corresponding age-specific template, using the paired T1w image as an intermediate reference. The workflow is illustrated in Fig. 2a. Preprocessed T1w images ($S_{\mathrm{orig},i}$, $i \in [1, \mathrm{n_g}]$; n_g is the sample size of each age group g) were registered to the age-specific template using the symmetric diffeomorphic normalization (SyN) algorithm[41] in Advanced Normalization Tools[42] (ANTs, https://github.com/ANTsX/ANTsPy) with default parameters. The resulting transformations were then applied to warp both the vessel segmentations ($V_{\mathrm{orig},i}$; nearest-neighbour interpolation) and the MRA images ($M_{\mathrm{orig},i}$; B-spline interpolation) into the age-specific template space. All templates were defined at an isotropic resolution of 0.5 mm³ to preserve vascular detail.

Because nonlinear warping may cause partial loss of small vessels, a re-segmentation step was introduced following Guo et al.[13]. The original, non-brain-extracted MRA images were first warped into the age-specific space to retain high-intensity vessel signals near the brain surface that may be removed during

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skull-stripping. Vessel structures were then re-segmented in the template space using the CereVessPro[13] model. The combined vessel mask was generated by combining the warped original segmentation with the re-segmented result, restoring vessel voxels lost during registration. Finally, a 5-mm dilated, age-specific brain mask was applied to each vessel map to remove common mis-segmented structures (e.g., optic nerve, cranial bone marrow) while retaining marginally displaced vessels and major arterial landmarks such as the internal carotid and basilar arteries, yielding the final vessel mask (V_i). This dilation was chosen based on anatomical considerations: major intracranial arteries near the brain boundary typically have diameters of approximately 3–4 mm[45,46], whereas the minimal distance from the cortex to cranial bone marrow is around 5 mm[47–49]. Therefore, a 5-mm dilation is sufficient to retain major intracranial vessels while minimizing contamination from extracranial structures.

Vascular refinement

Although T1w-guided structural alignment provides anatomically consistent registration, it is insufficient for fine-grained vascular correspondence because T1w images lack vascular landmarks and may introduce misalignment from the MRA-T1w co-registration. To address this, we implemented an additional vascular refinement. This refinement aimed to establish voxel-wise vascular correspondence across subjects by explicitly aligning vascular structures in a group-wise manner, thereby improving the anatomical fidelity of downstream probabilistic vessel maps. As illustrated in Fig.2b, the vascular refinement consisted of three sequential steps: (1) selection of a representative subject within each age group; (2) alignment of individual vessel to the representative mask; and (3) iterative group-wise map refinement.

Within each age group, one representative vessel mask was selected as the initial registration target using the strategy of Huck et al.[43]; the subject whose vessel map showed the smallest difference from the initial group-average served as the prototype:

$$\hat{\imath} = \arg\min_{i} D(V_i, A), \tag{1}$$

where D estimates the difference between individual vessel mask V_i obtained after structural alignment ($i \in [1, n_s]$; n_s is the sample size of the subset with full brain coverage) and the FOV-weighted mean vessel probability map (A) after structural alignment. D is computed as:

$$D(V_i, A) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \left| \frac{V_i(x) - A(x)}{\sigma(x) + \varepsilon} \right|, \tag{2}$$

where $\sigma(x)$ denotes the standard deviation across all subjects within corresponding FOV coverage at each voxel x, $\epsilon = 1 \times 10^{-6}$ is a small constant to avoid division by zero, $|\cdot|$ represents the absolute value operator, Ω denotes the whole-brain voxel set, and $|\Omega|$ is the number of voxels in Ω. Subsequently, each individual vessel mask is aligned to the representative mask $V_{\mathrm{R}} = V_{\hat{\imath}}$ via a transformation $\phi_{\mathrm{R},i}$, yielding the warped vessel $V_{\mathrm{R},i} = \phi_{\mathrm{R},i}(V_i)$, where $\phi_{\mathrm{R},i}$ is estimated by minimizing the mean squared error.

Next, a refined group-wise vessel probability map was generated using an established iterative procedure[44-46]. At each iteration k ($k \in [0, K]$), the transformation $\phi_{\mathrm{A}\to\mathrm{V},i,k}$ from the map A_k to each individual vessel $V_{\mathrm{R},i}$ was estimated with a cost function of cross-correlation (CC) at iteration. The estimation was initialized from the transformation $\phi_{\mathrm{V}\to\mathrm{A},i,k-1}$ (i.e., the inverse transformation of the vessel $V_{\mathrm{R},i}$, on the iteration $k-1$); for initialization, the identity transform was used as the initial

transformation $\phi_{\mathrm{V}\to\mathrm{A},i,0}$ and the initial map A_0 was set as the mean vessel probability from the previous step. The residual deformation was then calculated by averaging transformation of iteration k:

$$\bar{\phi}_{\mathrm{A}\to\mathrm{V},k} = \frac{1}{n_s}\sum_{i=1}^{n_s} \phi_{\mathrm{A}\to\mathrm{V},i,k}\,. \tag{3}$$

Subsequently, the transformation from each vessel $V_{\mathrm{R},i}$ was updated by the composition of the inverse of the current estimated transformation and the mean residual deformation from Eq. (3):

$$\phi_{\mathrm{V}\to\mathrm{A},i,k} = \left(\phi_{\mathrm{A}\to\mathrm{V},i,k}\right)^{-1} \circ \bar{\phi}_{\mathrm{A}\to\mathrm{V},k}, \tag{4}$$

and the next-iteration map A_{k+1} was updated by averaging all warped vessels:

$$A_{k+1} = \frac{1}{n_s}\sum_{i=1}^{n_s} \phi_{\mathrm{V}\to\mathrm{A},i,k}\left(V_{R,i}\right). \tag{5}$$

The procedure iteratively alternated between updating the deformation fields and refining the map (Eq. (3)-(5)) until convergence, which was defined as the root-mean-square magnitude of the residual deformation falling below a predefined threshold[45]. For implementation, we used Automatic Nonlinear Image Matching and Anatomical Labelling (ANIMAL)[47] to perform the non-linear registration within a subset of subjects with complete vessel coverage to prevent biases introduced by limited FOV. The algorithm employed multiscale refinement with progressively decreasing step sizes (16, 8, 4, 2 mm) and iteration counts (4, 4, 4, 2), totalling 14 iterations. This multiscale refinement strategy enabled the iterative map to progressively capture the stable and anatomically consistent vessel patterns shared across subjects within each age group. Finally, each individual vessel $V_{R,i}$ was nonlinearly registered to the iterative group map A_K using CC as cost function, yielding final registered individual vessel $V_{\mathrm{F},i} = \phi_{\mathrm{F},i}(V_{\mathrm{R},i})$ ($i \in [1,n]$; n is the sample size) and final vessel probability map $A_{\mathrm{F}}(x) = \frac{\sum_{i=1}^{n}(V_{\mathrm{F},i}(x))}{\sum_{i=1}^{n}(FOV_{\mathrm{F},i}(x))}$, where $FOV_{\mathrm{F},i}(x)$ is the corresponding warped binary mask indicating the FOV of subject i. To account for variations in FOV across datasets, registration was also restricted to the spatially overlapping region. All registrations underwent visual inspection to ensure anatomical plausibility and alignment quality.

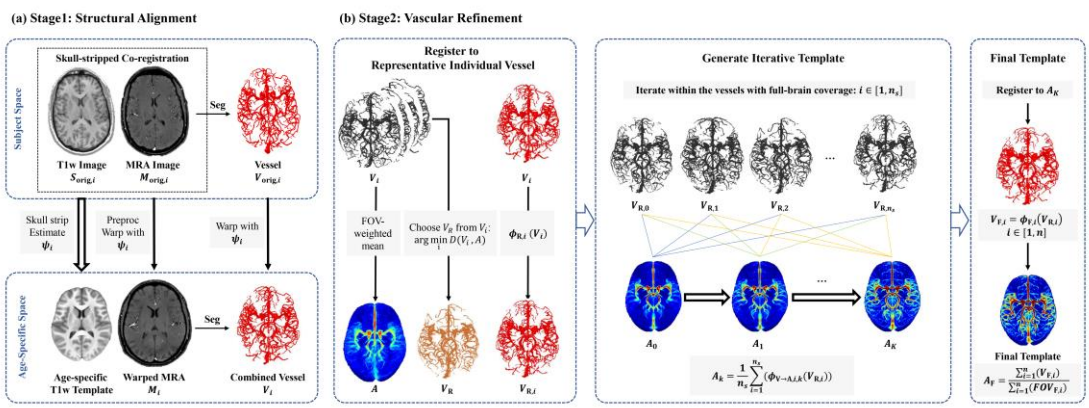


Figure 2. Schematic overview of the two-stage structural and vascular registration framework. (a) Structural alignment: T1w-guided registration to age-specific structural templates, followed by vessel re-

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segmentation in template space. **(b)** Vascular refinement: group-wise vessel map construction and nonlinear refinement of vascular features.

Age-specific cerebrovascular template generation

A comprehensive set of sixteen cerebrovascular templates was generated across four age groups (YO, YA, MA, OA). For each group, four types of maps were constructed—MRA, vessel probability, vessel density, vessel radius—resulting in 16 age-specific templates.

Vessel probability maps were constructed by incorporating each scan's FOV mask during template construction to account for inter-subject variation in anatomical coverage. Following the two-stage vascular alignment, vessel segmentations from all subjects were combined voxel-wise. The weighting at each voxel corresponded to the summed FOV masks across subjects. Mean vessel probability was then computed by dividing the voxel-wise sum of vessel segmentations by the corresponding FOV weights. To ensure robustness, only voxels covered by at least five subjects were considered valid in the final probability maps.

Templates construction for MRA intensity, vessel density and vessel radius followed the same transformation pipeline used during vascular alignment. Individual maps were warped into the appropriate age-specific template space using concatenated transformations to minimize cumulative interpolation error. MRA images were resampled using B-spline interpolation, and template intensities were computed as FOV-weighted voxel-wise averages. Vessel density maps were resampled using linear interpolation, and the mean vessel density map was generated by taking the FOV-weighted mean, excluding voxels with fewer than five valid samples. Radius maps were resampled using nearest-neighbour interpolation to preserve quantitative values, after which the mean radius map was generated by averaging all non-zero radius values at each voxel across subjects. For radius maps, a threshold requiring valid samples in over 30% of subjects was applied to balance the reliability of the radius estimates and spatial coverage.

To reduce residual boundary artefacts arising from segmentation or registration inaccuracies, an adapted brain mask was applied to each map. Particular attention was paid to ensure full preservation of the internal carotid artery (ICA), basilar artery (BA), and other circle of Willis (CoW) components, which are often truncated by conventional brain masks. CoW labels were generated using the CLAIM team's TopCoW model[48] (https://github.com/claim-berlin/TopCoW_2024_MRA_winning_solution) and subsequently warped into each age-specific space to obtain a population-level CoW mask. Following the TopCoW annotation protocol, the CoW region of interest (ROI) was defined as the 3D bounding box enclosing all CoW structures with additional padding. The final adapted brain masks for each age group were formed by integrating the structural template-derived mask with the population-based CoW mask, ensuring full coverage of all major CoW vessels in every template.

Data Records

The full set of age-specific cerebrovascular templates is publicly available at Figshare (https://doi.org/10.6084/m9.figshare.30948377.v1)[49]. All data are provided in NIfTI format with an isotropic resolution of 0.5mm^3 in the age-specific space. The files are named in [feature_name]_[age_group].nii.gz, providing templates in each age groups. The available feature categories are: "MRA" (intensity) and "Vessel" (probability) for anatomical reference, and "Density" and "Radius" (in mm) for quantitative analysis. Age groups are labelled as YO (youth), YA (young adults), MA (middle-aged adults), and OA (older adults). All volumes are stored as floating-point images to preserve quantitative precision. The corresponding age-specific T1w structural templates used for spatial

normalization are available in our previously published dataset[29] (https://doi.org/10.6084/m9.figshare.28093595.v1).

Technical Validation

Datasets

All datasets, as well as the outputs from each processing stage, including vessel segmentation and both anatomical and vascular alignment, were visually inspected to ensure quality for template generation. As shown in Fig.1b, scans were excluded if they exhibited excessive image noise, aliasing, motion artifacts, anatomical abnormalities, or poor registration quality. After completing this rigorous quality control pipeline, 1288 out of the initial 1337 datasets were included for further analysis.

Validation of the two-stage alignment

The impact of the two-stage alignment procedure is demonstrated using the OA cohort (Fig. 3a, b). After age-specific structural alignment alone, the maximum vessel probability in the group-averaged map reached 90.9%. Incorporating vascular alignment increased this value to 100%, indicating improved spatial correspondence of vascular structures across individuals. Visual inspection further confirmed sharper, more continuous vessel representations and enhanced alignment of both large arteries and smaller peripheral branches. To further assess the improvement at the individual level, we zoomed into the vascular structures within the CoW. After age-specific structural alignment alone, individual vessels still exhibited varying degrees of spatial deviation from the group-average vasculature. After vascular refinement, individual vessels showed improved correspondence with the group probability map, with more consistent alignment in both vessel trajectories and bifurcation locations.

To further quantify the improvement of vascular alignment, we compared the proposed method against three baseline pipelines: (1) T1-only structural registration (without vascular refinement); (2) an age-agnostic template construction pipeline; (3) a simplified MNI-based pipeline[20]. Validation was performed on a randomly selected subset of 20 OA subjects using a pairwise strategy, yielding 190 subject pairs. We assessed alignment using seven metrics covering overlap (Dice, Intersection over Union [IoU], and clDice[50]), surface distance (Average Symmetric Surface Distance [ASSD] and 95th percentile Hausdorff Distance [HD95]), and centerline-based distances, computed at both global vascular and Circle of Willis (CoW) scales. As shown in Fig. 3c, the proposed method consistently outperforms all baselines.

In addition, to simulate a real-world scenario where a new subject is registered to a pre-existing reference, we performed a leave-one-subject-out experiment. Each held-out subject (not involved in the iterative vascular refinement stage) was registered to a group template constructed from the remaining subjects. Alignment was quantified using soft Dice at both global and CoW levels. The results, shown in Fig. 3c, demonstrate that the proposed framework improves vascular alignment not only within the construction cohort but also for unseen subjects, supporting its practical utility for real-world applications.

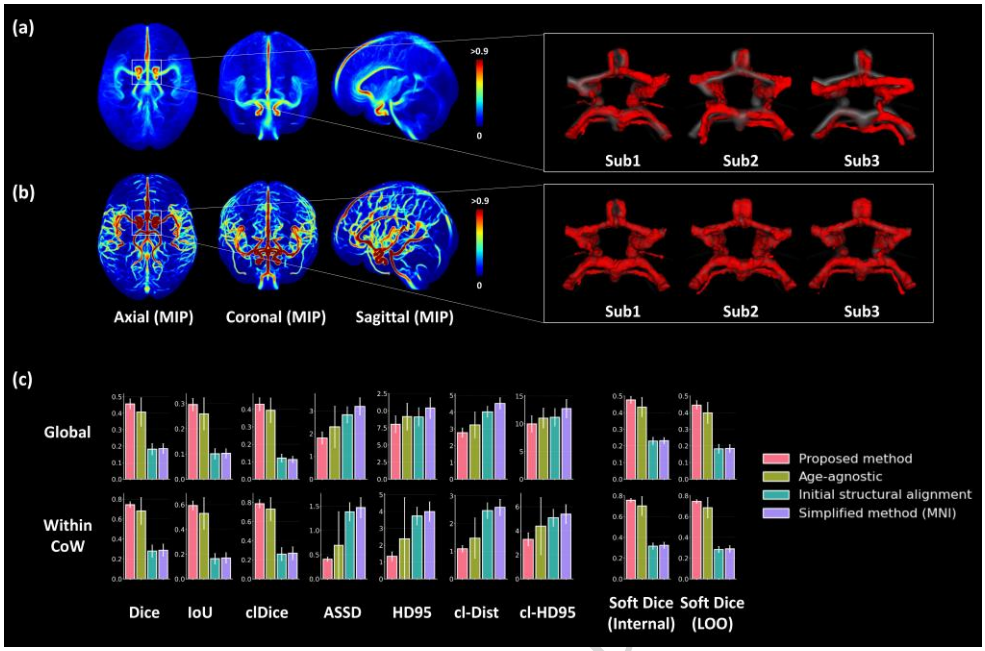


Figure 3. Effect of the two-stage alignment approach on vascular probability estimation in the older adults (OA) group. Maximum intensity projections (MIPs) of the mean vessel probability map following **(a)** initial structural alignment, and **(b)** subsequent vascular refinement. Zoomed-in views highlight the circle of Willis (CoW) region, showing vascular structures from three representative individual subjects (red) overlaid on the final vessel probability map (grayscale). The vascular refinement stage improves spatial concordance of major arterial structures across subjects. **(c)** Quantitative performance of proposed methods against three baseline pipelines at both the global level and within the CoW region.

Age-specific cerebrovascular template

Figure 4 presents the age-specific mean MRA templates and corresponding vessel probability maps for each age group. These maps provide essential vascular anatomical references, facilitating the localization of vascular structures and enabling the alignment of future vessel data to our age-specific template spaces. The mean MRA templates reveal a progressive enlargement of the ventricles across age groups, consistent with the structural changes in the corresponding T1w templates[29]. Moreover, several age-related morphological changes in the cerebral vasculature are observed, such as the increased tortuosity of the ICA with advancing age[15,51,52]. The vessel probability maps (Fig. 4b) demonstrate spatial relationship between the vascular structures and tissue structures visible on the T1w images, reflecting the established biological association between vascular architecture and cortical folding[53]. Additionally, maximum intensity projections (MIPs) of the mean vessel probability maps (Fig. 4c) demonstrate the overall alignment of vascular structures in each age group.

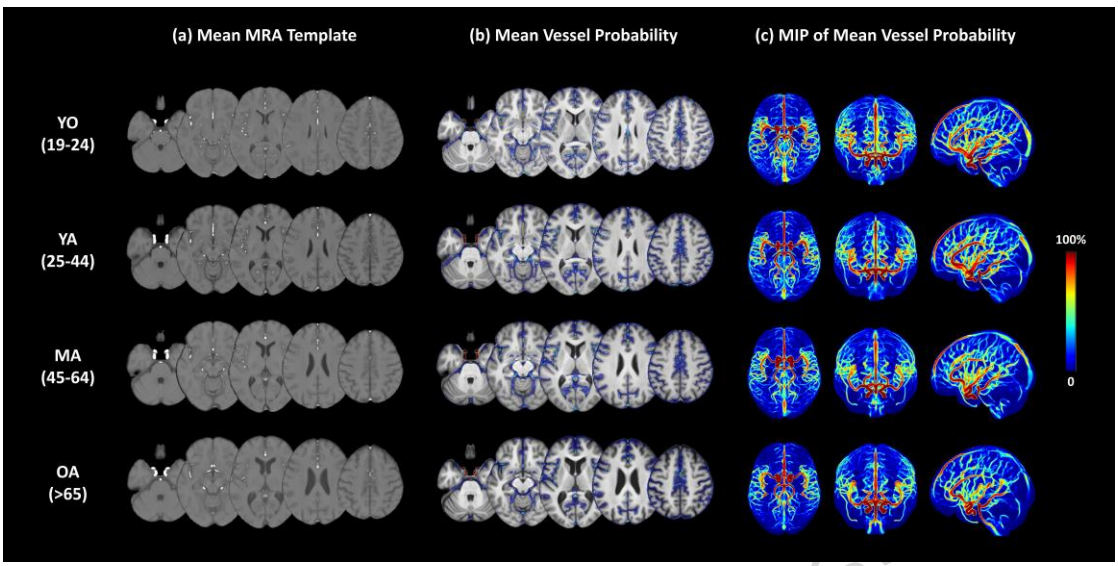


Figure 4. Age-specific cerebrovascular templates for vascular anatomical references. (a) Representative slices from the mean MRA templates for each age group. **(b)** Representative slices from the mean vessel probability maps overlaid on the corresponding T1w templates for each age group. **(c)** Three-dimensional maximum intensity projection (MIP) renderings of the mean vessel probability maps for each age group. Abbreviations: YO, youth; YA, young adults; MA, middle-aged adults; OA, older adults.

Figure 5 illustrates a clear age-related decline in overall cerebral vessel density[33], coupled with a concurrent increase within the CoW region[13], suggesting a general reduction in the small vessels with aging, alongside a compensatory enlargement in the major vessels.

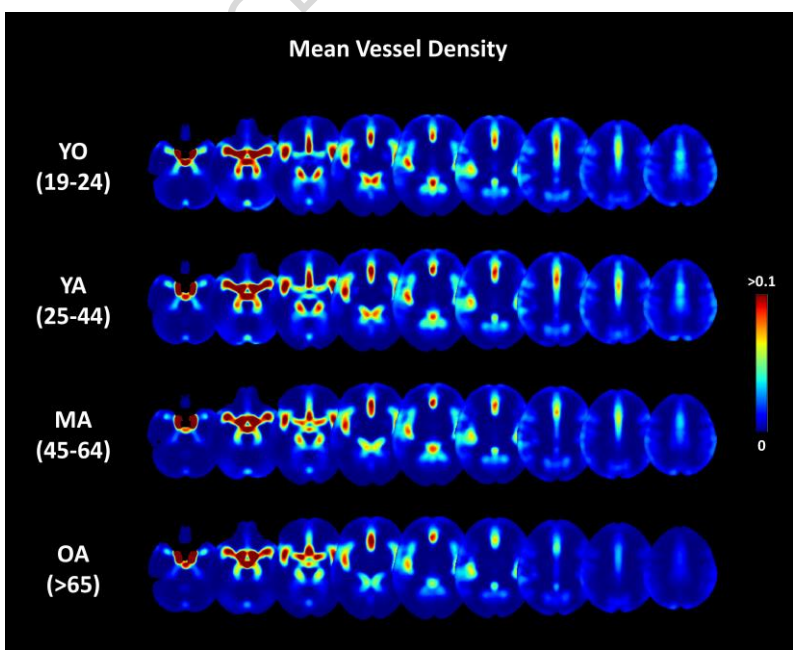


Figure 5. Age-specific vessel density maps. Selected slices from the mean vessel density maps are shown for each age group. Abbreviations: YO, youth; YA, young adults; MA, middle-aged adults; OA, older adults.

Figure 6 presents the overall distributions of mean vessel radius across age groups. To further quantify age-related changes in large vessels, the distribution of mean radius values within the CoW was analysed

using a public available CoW atlas from Guo et al[13]. A progressive rightward shift in the upper tail of the radius distributions is observed with increasing age, indicating systematic enlargement of large-vessel radii across age groups (Fig. 6b).

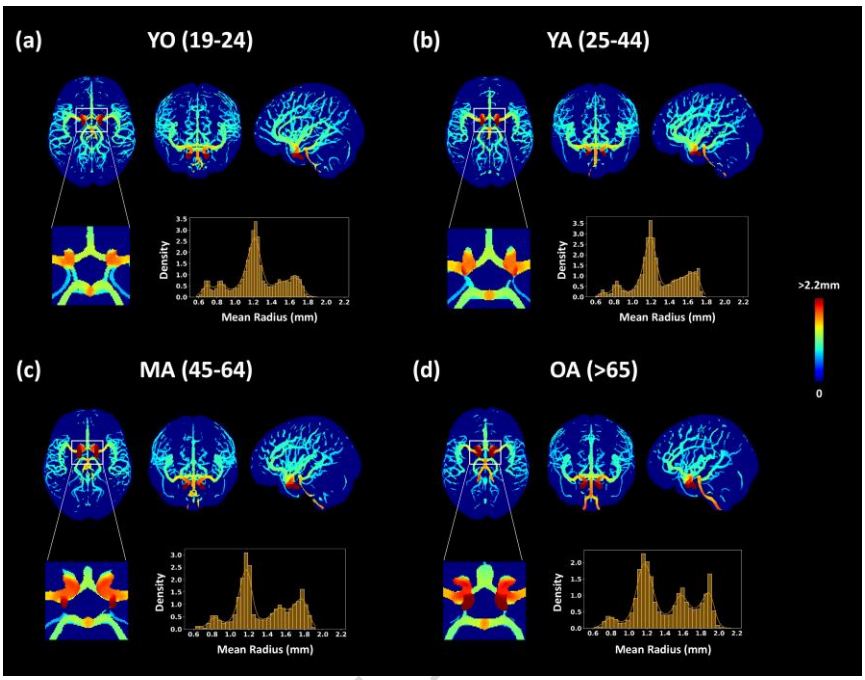


Figure 6. Age-specific vessel radius maps. Maximum intensity projections (MIPs) of the mean vessel radius maps are shown for each age group, along with representative local regions in the circle of Willis (CoW) and the corresponding histograms of vessel radius values: Panels correspond to (a) youth (YO), (b) young adults (YA), (c) middle-aged adults (MA), and (d) older adults (OA).

Validation of cerebrovascular density

Global cerebrovascular density was quantified to assess age-related variations. The mean density values were calculated as the mean density within the valid FOV, with region masks defined with a brain mask excluding dural sinuses and CoW regions in the standard space. To ensure data reliability, measurements for a region were considered valid only if the FOV coverage of the region mask exceeded 80% for each individual. After applying this coverage-based filtering criterion, a total of 567 subjects were included in this analysis, with relative percentage of each dataset within each age group shown in Fig. 7c, f. Age-related associations were assessed using Spearman correlation and linear regression analyses, with post hoc FDR-corrected t-tests used to evaluate pairwise group differences.

A clear age-related decline in global vessel density was observed (Fig. 7a, b), consistent with previously reported trends[9,13,33,34,54,55]. Mean global vascular density decreased progressively across age groups (YO: 0.0116; YA: 0.0116; MA: 0.0105; OA: 0.0099), with all inter-group comparisons reaching significance (P_{FDR} < 0.05) except between YO and YA. The age effect remained significant (p < 0.001) after include site as a categorical covariate in the regression model. Sex differences are summarized in Fig. 7d, e. Across all age groups, females showed significantly higher vessel density than males (P_{FDR}< 0.05), in line with previous reports of higher microvessel density[56] and higher CBF[29,57,58] in females.

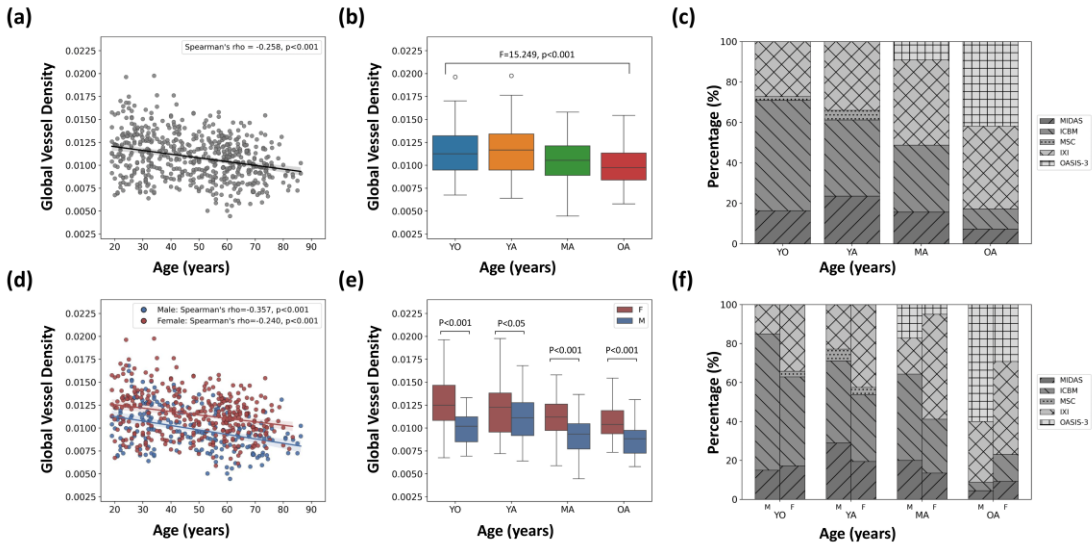


Figure 7. Age-related changes in global cerebrovascular vessel density. (a, b) Scatter plots and box plots illustrating mean vessel density across age groups. (c) Percentage contribution of individual datasets. (d, e) Scatter plots and box plots comparing mean vessel density between males and females. (f) Sex-specific percentage contribution of individual datasets. Linear regression fits with 95% confidence intervals (CIs) are shown. Abbreviations: YO, youth; YA, young adults; MA, middle-aged adults; OA, older adults.

Validation of cerebrovascular radius

Quantitative radius measurements were evaluated using manually defined regions-of-interest (ROIs) placed on major arteries within the mean MRA templates. These ROIs were then propagated to each subject's native radius map to calculate average arterial radii. The placement of ROI followed the criteria described by Groenheide et al.[59]: the A1 segment of the anterior cerebral artery (ACA), M1 segment of the MCA, and P1 segment of the PCA were measured at their midpoints; the ICAs were measured 5 mm below the carotid top; the BA was measured 5 mm inferior to its bifurcation. All ROIs were centered precisely on the vessel centerline to ensure sampling consistency. Measurement locations are illustrated in Fig. 8a. For bilateral arteries (ACA, MCA, PCA, ICA), left and right measurements were averaged.

To characterize normative arterial geometry, mean radii across all subjects were computed for each arterial segment (Fig. 8b). Due to incomplete field-of-view coverage of the Circle of Willis in a small number of subjects, the effective sample size varied slightly across arterial segments (ICA/MCA/PCA: n = 1287; BA: n = 1286; ACA: n = 1285). As a result, the ICA exhibited a mean radius of 1.70 mm, closely matching the pooled value of 1.7 mm reported for the C7 segment by Mirza et al.[60]. The BA demonstrated a mean radius of 1.57 mm, consistent with the 1.59 mm reported by Smoker et al.[61]. The mean ACA-A1 radius was 1.18 mm (previously reported: 1.1 mm[14]), the MCA-M1 radius was 1.44 mm (slightly higher than the 1.3 mm reported by Mirza et al.[60], but more close with 1.495 mm derived from postmortem measurements[62] and 7T vessel wall data[63]), and the PCA-P1 radius was 1.18 mm (previously 1.05 mm[14]). Minor discrepancies between studies likely reflect variations in manually chosen measurement sites and subjective boundary delineation, both of which influence cross-sectional diameter estimation. Nevertheless, our measurements, derived from segmented TOF-MRA, can be affected by segmentation accuracy and remain susceptible to partial volume effects and the finite spatial resolution of the imaging data[64]. As a result, these estimates should not be interpreted as direct measurements of true anatomical vessel diameters and should instead be interpreted with caution. Sex-related differences in vessel calibre were also examined. Across all arterial segments, males displayed significantly larger radii than females

(Fig. 8c), a finding consistent with previous studies[59,65]. All comparisons reaching statistical significance (P_{FDR} < 0.001). Corresponding effect sizes (Cohen's d) are shown above each comparison, further supporting robust and consistent sex differences in intracranial arterial radius.

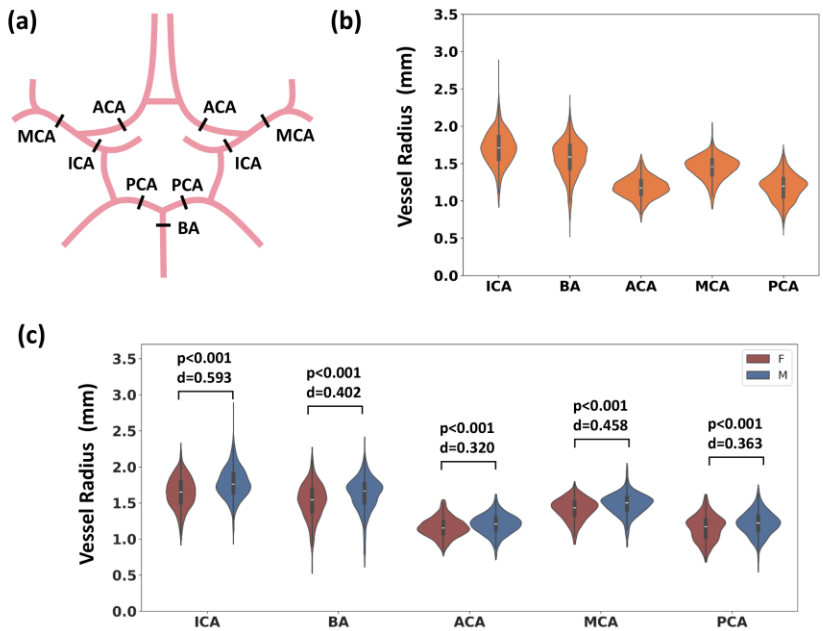


Figure 8. Validation of cerebrovascular radius measurements. (a) Schematic illustration of the anatomical locations of manually defined regions of interest (ROIs) used for vessel radius quantification, corresponding to major cerebral arteries. (b) Violin plots showing the distribution of vessel radius measurements across the five arterial segments. (c) Sex-specific comparisons of vessel radius. Abbreviations: ACA, anterior cerebral artery; MCA, middle cerebral artery; ICA, internal carotid artery; PCA, posterior cerebral artery; BA, basilar artery.

Age trends in cerebrovascular radius

We further examined age-related trends in radius across arterial segments. Analysis of age-related trends in vessel radius revealed distinct patterns across arterial segments, as shown in Fig. 9. The ICA demonstrated a progressive increase in radius with advancing age, consistent with previous reports[65-69] and potentially reflecting age-related compensatory dilation or vascular remodeling. A similar age-related enlargement was observed in the MCA, aligning with findings from Mouches et al.[70]. In contrast, ACA exhibited a decreasing radius with age, a pattern that agrees with earlier observations[65,69]and may indicate differential susceptibility of anterior circulation branches to aging processes. The age effect remained significant ($p < 0.001$) after including site as a categorical covariate in the regression models. No clear age-dependent trends were detected in the remaining arterial segments. Sex differences persisted across the lifespan: males consistently exhibited larger arterial radii than females in all examined segments and across all age groups (Fig. 9), in agreement with prior studies[59].

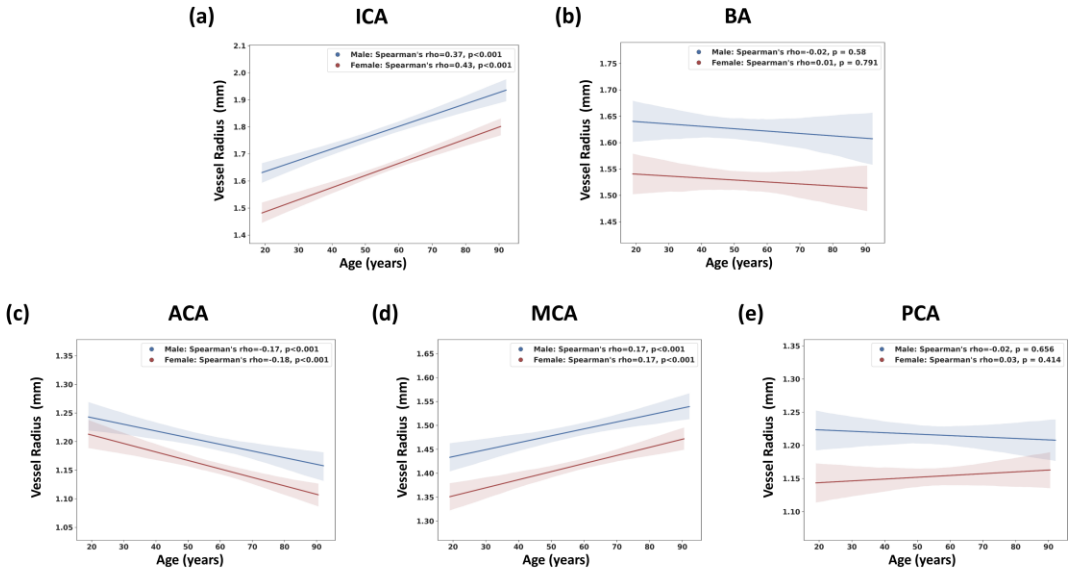


Figure 9. Age-related trends in vessel radius across major cerebral arteries. Linear regressions of vessel radius versus age for (a) internal carotid artery (ICA), (b) basilar artery (BA), (c) anterior cerebral artery (ACA), (d) middle cerebral artery (MCA), and (e) posterior cerebral artery (PCA). Blue and red lines denote male and female participants, respectively, with shaded bands indicating the corresponding 95% confidence intervals (CIs).

Data Availability

The full set of age-specific cerebrovascular templates is openly available at (https://doi.org/10.6084/m9.figshare.30948377.v1)[49].

Code Availability

The vessel segmentation model used in this work was developed by Guo et al.[13] , with both the source code and trained models publicly accessible at [https://github.com/YingChen7/HQA_Cerebrovasculature]. For the iterative registration procedures, we utilized tools developed by Fonov et al.[44-46], which are also openly available at [https://github.com/NIST-MNI/nist_mni_pipelines].

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Author contributions

C.X: Study design, image processing, data analysis, drafting and revision of the manuscript critically. Z.M.: Image processing, data analysis, drafting and revision of the manuscript critically. H.Z.: Image processing and revision of the manuscript. R.L.: Image processing and revision of the manuscript. L.C.: Revision of the manuscript critically. Z.L.: Revision of the manuscript critically. Y.Liu: Revision of the manuscript critically. Y.Li: Study design, interpretation of the data, drafting and revision of the manuscript critically.

Competing interests

The authors declare no competing interests.

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